FIITJEE

ALL INDIA TEST SERIES

FULL TEST –X

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 20-09-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C

Sol. When speed is maximum, then $F_{electrostatics} = F_{friction}$

$\Rightarrow \frac{Kq^2}{r^2} = \mu mg \Rightarrow r = 3m$

$\therefore$ C is correct

$\frac{Kq^2}{1} = \frac{Kq^2}{3} + 2\times\frac{1}{2}mv^2 + \mu mg(3-1) \Rightarrow v = 2$ m/s

$\therefore$ A is correct

The particle will again come to rest at a finite separation where the potential energy will not be zero. $\therefore$ D is incorrect.

2. A, B, C, D

Sol. $R \propto A^{1/3}$

As nucleus has equal no of protons and neutrons, so A = 2z.

$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{z_1}{z_2}\right)^{1/3}$

$\Rightarrow \frac{z_1}{z_2} = \left(\frac{R_1}{R_2}\right)^3 = \frac{27}{8}$

Radius of the 1st Bohr's orbit $r \propto 1/z$

$\therefore \frac{r_1}{r_2} = \frac{z_2}{z_1} = \frac{8}{27}$

Velocity of e^- in the first Bohr's orbit $v \propto z$

$\therefore \frac{v_1}{v_2} = \frac{z_1}{z_2} = \frac{27}{8}$

Ionisation energy $E = 13.6z^2\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right]$

$\frac{E_1}{E_2} = \frac{z_1^2}{z_2^2} = \left(\frac{27}{8}\right)^2 = \frac{729}{64}$

Angular momentum $L = \frac{nh}{2\pi}$

$\therefore \frac{L_1}{L_2} = 1$

3. A, B, D

Sol. Maximum Kinetic energy of β_1^- = {(197.968233 – 197.966760) × 931.5} – 1.088 = 0.284 MeV

Maximum kinetic energy of β_2^- = {(197.968233 – 197.966760) × 931.5} – 0.412 = 0.960 MeV

So, option A and B are correct.

Energy associated with γ_1 is 1.088 MeV. So option (C) is incorrect.

From the nuclear energy level diagram of $^{198}_{80}Hg$ it is clear that energy released during emission of γ_3 is (1.088 – 0.412) MeV = 0.676 MeV.

4. A, B, C

Sol. For rotational equilibrium of cylinder A: $f_A = f_{AB}$

For rotational equilibrium of cylinder B: $f_B = f_{AB}$

For translational equilibrium of (A + B)

$f_A + f_B = (M_A + M_B)\, g \sin\theta$

$\Rightarrow f_A = f_B = f_{AB} = \left(\frac{M_A + M_B}{2}\right) g\sin\theta$

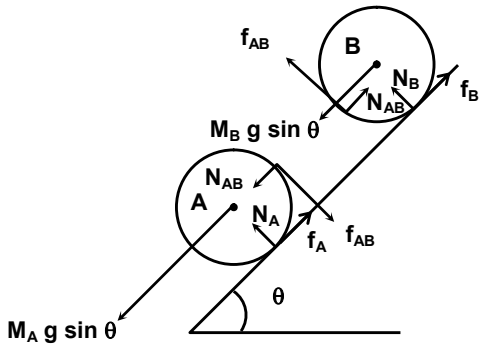
For A to be in equilibrium

$N_{AB} + M_A g \sin\theta = \left(\frac{M_A + M_B}{2}\right) g\sin\theta$

$\Rightarrow N_{AB} = \left(\frac{M_B - M_A}{2}\right) g\sin\theta$

For 2 cylinders to be in contact $N_{AB} > 0$

$\Rightarrow M_B > M_A$



5. A, B, C, D

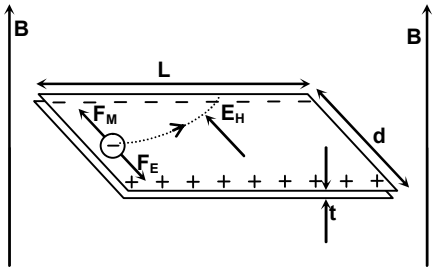
Sol. $P = K\frac{E}{V}$

$\therefore PV = KE$

Also for ideal gas PV = nRT.

6. B, D

Sol. Due to motion of electron in magnetic field it will experience magnetic force (F_m). As a result of this they will accumulate on one side leaving behind positive charges on the other as shown in the figure. This results in development of an electric field E_H, and will finally become constant when $F_E = F_M$. At this instant $V = E_H d$



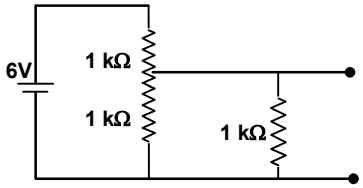
$$\therefore \quad E_H = \frac{V}{d} \Rightarrow E_H = \frac{3\times10^{-6}}{10^{-2}} = 3\times10^{-4}\ \text{V/m}$$

Electric force on e^- is $F_E = qE_H = 1.6 \times 10^{-19} \times 3 \times 10^{-4} = 4.8 \times 10^{-23}$ N

Drift velocity of the e^- is $\frac{E_H}{B} = \frac{3\times10^{-4}}{2} = 1.5\times10^{-4}$ m/s

7. A, C, D

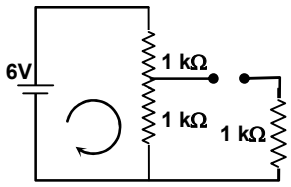
Sol. Before switch S_2 is opened, the circuit is as follows:



$$\therefore\ i_{battery} = \frac{6}{\frac{3}{2}\times10^3} = 4\ \text{mA}$$

$i_{R'} = 2\text{mA} \qquad \therefore\ V_0 = i_{R'} \times R' = 2V$

For t > 0 and at steady state, $\Delta V_C = V_1 = 3V$



$$\therefore\ \frac{V_1}{V_0} = \frac{3}{2}$$

$\therefore$ option (A) is correct and (B) is incorrect.

Time constant of the circuit when the capacitor is getting charged is

$$\tau = R_{eq}C = \frac{3}{2}\times10^3 \times 8\times10^{-3} = 12\,\text{s}$$

$$\therefore\ V_C = 3\left[1-e^{-\frac{t}{12}}\right],\ q = CV = 24\left[1-e^{-\frac{t}{12}}\right]$$

$$i_C = \frac{dq}{dt} = 24\times\frac{1}{12}e^{-\frac{t}{12}} = 2e^{-\frac{t}{12}}(\text{mA})$$

$P_C = V_C i_C$

$$P_C = 6\left[1-e^{-\frac{t}{12}}\right]e^{-\frac{t}{12}} = 6(1-x)x$$

$\therefore P_C$ will be maximum when x = 1/2

i.e., $e^{-\frac{t}{12}} = \frac{1}{2} \Rightarrow -\frac{t}{12} = -\ell n(2) \Rightarrow t = 12\ \ell n(2)$

$$\therefore\ (P_C)_{max} = 6\left(1-\frac{1}{2}\right)\times\frac{1}{2} = 1.5\,\text{mW}$$

$\therefore$ option (C) is correct

Total heat dissipated across R′ after opening S_2,

$$H = \int_0^\infty i_C^2 R'dt = \int_0^\infty 4\times10^{-6}\times e^{-\frac{t}{6}}\times10^3 dt$$

$= 4 \times 10^{-3} \int_0^{\infty} e^{-\frac{t}{6}} dt = 24$ mJ

$\therefore$ option (D) is correct

8. A, B, C, D

Sol. For refraction at AB: 1 sin 45° = $\sqrt{2}$ sin $r_1 \Rightarrow r_1 = 30°$

$\therefore$ Ray within prism is parallel to BC

$\therefore$ D is correct

For refraction at AC: $\sqrt{2}$ sin 30° = 1 sin $i_2 \Rightarrow i_2 = 45°$

Deviation is $\delta = (i_1 + i_2) - A = 30°$, and the ray after emerging from prism will be incident on the concave mirror parallel to its principal axis (from geometry). So, the final image formed after reflection from mirror will be at its focus i.e. 15 cm from its pole and the image will be real, inverted and diminished.

$\therefore$ B is correct.

Y-coordinate of the final image is 15 sin 15° = $\frac{15(\sqrt{3}-1)}{2\sqrt{2}}$ cm.

$\therefore$ C is also correct.

9. D

10. B

Sol. **(for Q.9–Q.10):**

$$T = \frac{2\pi}{\sqrt{GM}} r_0^{3/2}$$

$$E_1 = \frac{GMm(3R^2 - r_L^2)}{2R^3} - \frac{GMm}{r_0}$$

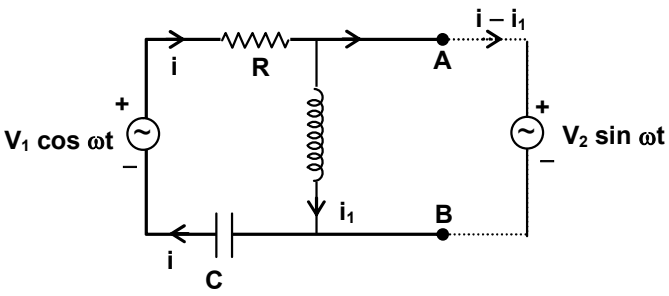
$$E_2 = \frac{GMm}{2r_0}$$

11. C

12. A

Sol. **(for Q.11–Q.12):**

Using KVL in outer loop



$v_0 \cos \omega t - iR - v_0 \sin \omega t - \frac{q}{C} = 0$

$\Rightarrow v_0(\cos \omega t - \sin \omega t) - iR - \frac{q}{C} = 0$

$\Rightarrow \sqrt{2} v_0 \cos\left(\omega t + \frac{\pi}{4}\right) - iR - \frac{q}{C} = 0$

$\therefore i = \frac{\sqrt{2} v_0}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \cos\left(\omega t + \frac{\pi}{4} + \phi\right) = i_0 \cos\left(\omega t + \frac{\pi}{4} + \phi\right)$

$\phi = \tan^{-1}\left(\frac{x_C}{R}\right) = \tan^{-1}\left(\frac{1}{\omega CR}\right)$

$$v_R = iR = \frac{\sqrt{2}v_0 R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}} \cos\left(\omega t + \frac{\pi}{4} + \phi\right)$$

Phase difference between i and $v_1 \cos \omega t$ is $\phi' = \frac{\pi}{4} + \phi$

Case 1: when $\omega = 0$ then $v_1 = v_0$, $v_2 = 0$, $i_0 = 0$, $\phi = \pi/2$

$\therefore \ \phi' = \frac{3\pi}{4}$

Case 2: when $\omega = \frac{R}{L} = \frac{1}{RC}$, then $\omega = \frac{1}{\sqrt{LC}}$, $i_0 = \frac{v_0}{R}$, $\phi = \pi/4$

$\therefore \ \phi' = \frac{\pi}{2}$

Case 3: when $\omega = \frac{2R}{L} = \frac{1}{RC}$ then $i_0 = \frac{v_0}{R}$, $\phi = \pi/4$

$\therefore \ \phi' = \frac{\pi}{2}$

Case 4: when $\omega = \frac{R}{L} = \frac{1}{\sqrt{3}RC}$ then $i_0 = \frac{v_0}{\sqrt{2}R}$, $\phi = \pi/3$

$\therefore \ \phi' = \frac{7\pi}{12}$

SECTION – C

13. 00001.50

Sol. $T \propto P^a d^b E^c$

$\Rightarrow [M^0L^0T] = [ML^{-1}T^{-2}]^a [ML^{-3}]^b [ML^2T^{-2}]^c$

Equating exponents of M and T, we get

$a + b + c = 0$ and $-2a - 2c = 1$

$\Rightarrow a + c = -\frac{1}{2}$

$\Rightarrow b = \frac{1}{2}$

Hence $a + 4b + c = 3/2 = 1.5$

14. 00003.25

Sol. $r_{AB} = \int_0^1 \lambda dx = \int_0^1 (1 + 2x)dx = 2\Omega$

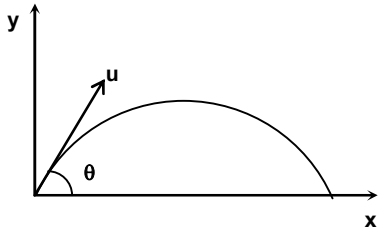
$r_{AC} = \int_0^{0.5} (1 + 2x)dx = \frac{3}{4}\Omega$

Potential difference between X and Y = 4V = potential difference between A and C when $i_G = 0$

$\therefore \ 4 = \frac{28}{(r+2)} \times \frac{3}{4} \Rightarrow r = 3.25\ \Omega$

15. 00195.00

Sol. Coordinates of the ball at t = t is



$x = u\cos\theta t$

$y = u\sin\theta t - \frac{1}{2}gt^2$

and time of flight is $\frac{2u\sin\theta}{g}$

Area under trajectory is

$$A = \int_0^x ydx = \int_0^{\frac{2u\sin\theta}{g}} \left(u\sin\theta t - \frac{gt^2}{2}\right) u\cos\theta dt$$

$$\therefore \quad A = \frac{2u^4}{3g^2}\sin^3\theta\cos\theta$$

For A to be maximum, $\frac{dA}{d\theta} = 0$

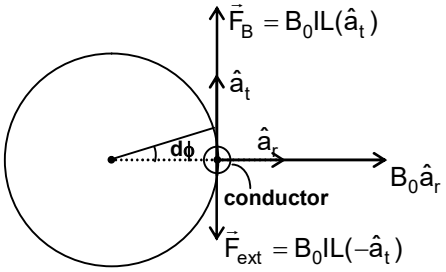
$\Rightarrow \quad \theta = 60°$

$$\therefore \quad A_{max} = \frac{\sqrt{3}u^4}{8g^2} = 194.85 \approx 195 \text{ m}^2$$

16. 00000.50

Sol. $\vec{F}_B = I(\vec{L}\times\vec{B}) = I(L\hat{K}\times B_0\hat{a}_r) = B_0IL\hat{a}_t$

$\vec{F}_{external} = B_0IL\left(-\hat{a}_t\right)$



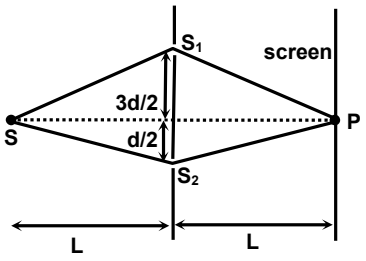
Work done by external agent to turn the conductor by 2π radian

$$W = \int_0^{2\pi} B_0IL\left(-\hat{a}_t\right)\cdot rd\phi\hat{a}_t = -2\pi rB_0IL$$

$P = 2\pi rB_0ILN = 0.5$ W

17. 04800.00

Sol. $\frac{3}{4}I_{max} = I_{max}\cos^2\frac{\phi}{2} \Rightarrow \cos\frac{\phi}{2} = \frac{\sqrt{3}}{2}$



$\Rightarrow \frac{\phi}{2} = \frac{\pi}{6} \Rightarrow \phi = \pi/3$

$\phi = \frac{2\pi}{\lambda}\Delta x = \frac{\pi}{3} \Rightarrow \Delta x = \frac{\lambda}{6}$

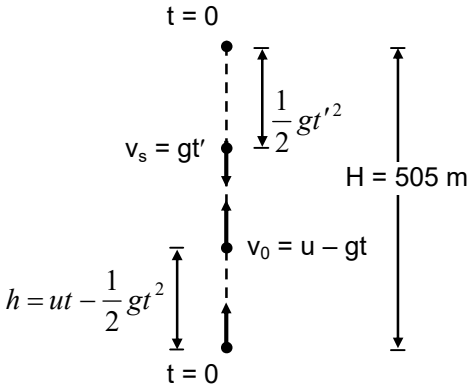
Given d = 200 μm = 2×10^{-4} m

$$2\left[\sqrt{L^2 + \frac{9d^2}{4}} - \sqrt{L^2 + \frac{d^2}{4}}\right] = \frac{\lambda}{6}$$

$$\Rightarrow \frac{2d^2}{L} = \frac{\lambda}{6} \quad \Rightarrow \lambda = \frac{12d^2}{L} = \frac{12\times4\times10^{-8}}{1} = 4800\text{Å}$$

18. 01800.00

Sol. Let t′ be the time at which the source emits sound wave which is detected at t = 5s. Then,



$$H - \frac{1}{2}gt'^2 - h = v(t - t')$$

or $$505 - \frac{1}{2}gt'^2 - \left(ut - \frac{1}{2}gt^2\right) = v(t - t')$$

using t = 5s ; g = 10 ms^{-2} ; v = 300 ms^{-1}, we get

$505 - 5t'^2 - 125 = 300\,(5 - t')$

or $t'^2 - 60\,t' + 224 = 0$

$\Rightarrow$ t′ = 4s

Using Doppler's formula

$$f = f_0\left(\frac{v + v_0}{v - v_s}\right)$$

here $v_s = gt' = (10)(4) = 40\ ms^{-1}$

$v_0 = u - gt = 50 - (10)(5) = 0$

$\therefore$ $$f = 1560\left(\frac{300}{300 - 40}\right) = 1800\ Hz$$

Chemistry

PART – II

SECTION – A

19. A, B

Sol. Both gases will have same pressure

$P_A = P_B$

Chamber 1: $P_AV_A = n_ART_A$ … (i)

Chamber 2: $P_AV_B = n_BRT_B$ … (ii)

Eqn. (i) ÷ (ii)

$$\frac{P_AV_A}{P_BV_B} = \frac{n_ART_A}{n_BRT_B}$$

$$3 = \frac{1}{3}\cdot\frac{T_A}{T_B}$$

$T_A = 9T_B$

$T_A > T_B$

Heat will not flow 1 to 2 because chamber is insulating and piston is non conducting.

20. A, B, C, D

Sol. Borax bead test is given by Cu, Fe, Cr, Mn, Co and Ni metals.

21. A, C

Sol. $As_2S_3 + (NH_4)_2S_2 \longrightarrow As_2S_5 + (NH_4)_2S$

$As_2S_5 + 3(NH_4)_2S \longrightarrow \underset{\text{ammonium thioarsenate}}{2(NH_4)_3AsS_4}$

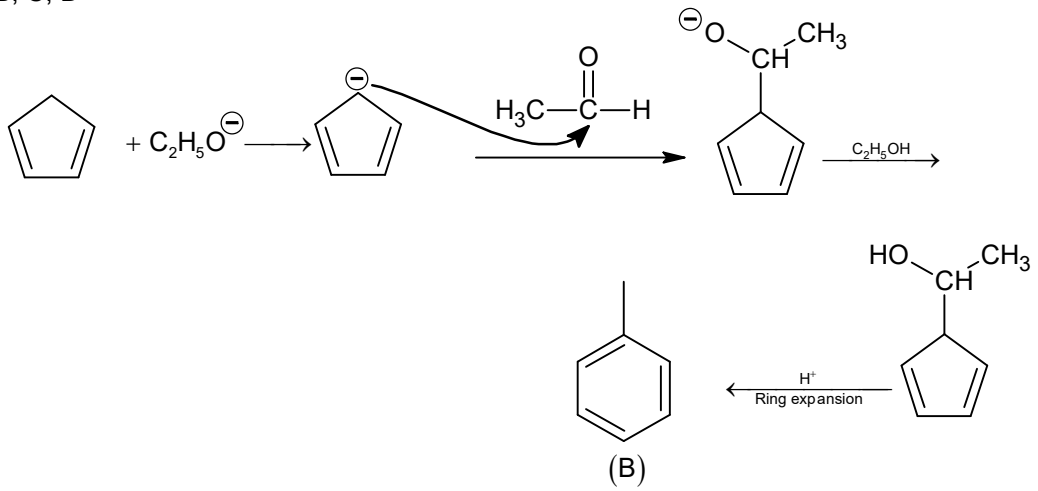
$NH_4Cl + NaOH \longrightarrow NaCl + H_2O + NH_3(g)$

22. B, C, D

Sol. Compound with white colour – $PbCl_2$, $Al(OH)_3$, $BaCO_3$, $SrCO_3$, $Mg(NH_4)HPO_4$.

23. B, C, D

Sol.



(B)

24. A, C, D

Sol. (i) Process AB → V increases, T increase $\Delta T > 0$, $\Delta U > 0$

(ii) Process BC → V decreases, T decrease $\Delta T < 0$.

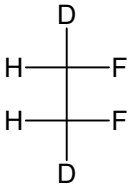
(iii) Process CD → pressure decreases, T decreases $\Delta T < 0$, $\Delta Q < 0$.

(iv) Process DA → pressure increase, $\Delta Q > 0$.

25. A, B, C

26. A, C

Sol. A and C are identical and have meso form.



(meso form)

27. C

28. B

Sol. **(for Q.27 – Q.28):**

Glyptal – Lacquers
Bakelite – Computer discs
Polystyrene – Radio cabinets
Urea-formaldehyde resin – Unbreakable cups, Laminated sheets

29. C

30. B

Sol. **(for Q.29 – Q.30):**

Dolomite – $MgCO_3.CaCO_3$
Limonite – $2Fe_2O_3.3H_2O$
Azurite – $Cu(OH)_2.2CuCO_3$
Alunite – $K_2SO_4.Al_2(SO_4)_3.2Al(OH)_3$

SECTION – C

31. 00001.25

Sol. Total number of correct statements are 5
a, b, c, e, h are correct. So x = 5
Total incorrect statements are 4
d, f, g, i are incorrect. So y = 4

$$\frac{x}{y}=\frac{5}{4}=1.25$$

32. 00018.75

Sol. a is edge length of unit cell, r is the radius of carbon atom. In diamond unit cell carbon atoms are present at C.C.P. and alternate tetrahedral void position.

$$\frac{a\sqrt{3}}{2}=4r \Rightarrow a\sqrt{3}=8r$$

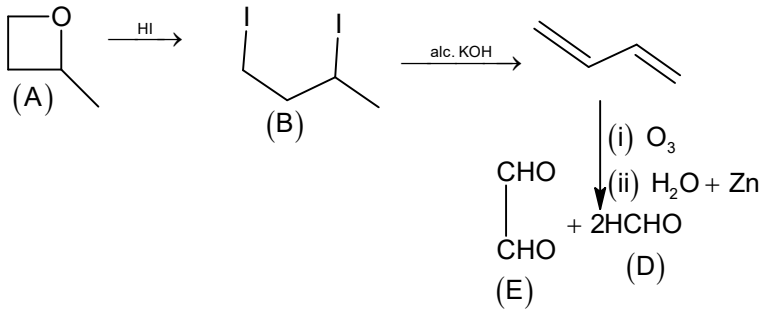
Net atoms present one face = 2

$$\% \text{ age of area occupied by atoms } =\frac{2\pi r^2}{a^2}=\frac{2\pi r^2}{\left(\frac{8r}{\sqrt{3}}\right)^2}\times 100$$

$$=\frac{2\pi r^2\times 100\times 3}{64r^2}=\frac{300\pi}{32}$$

33. 00001.50

Sol.



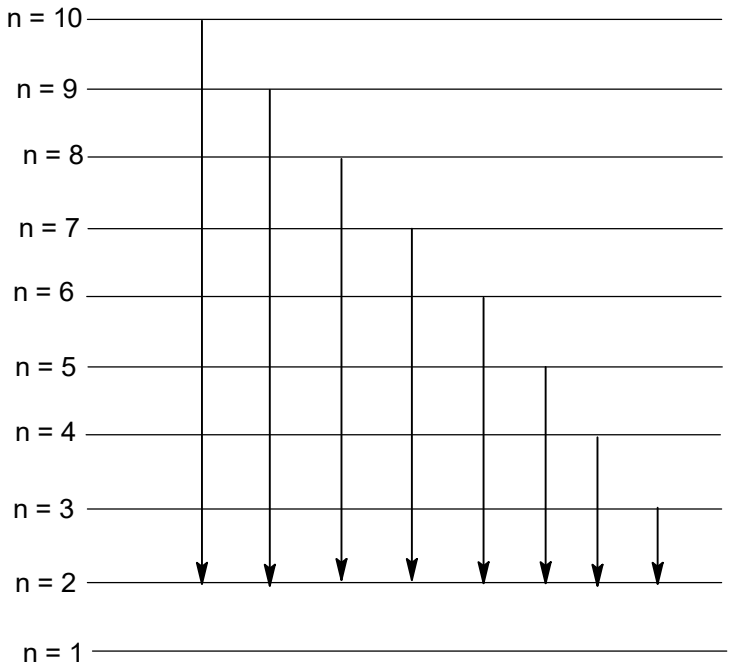
Number of atom in E = 6

Number of atom in D = 4

$\frac{6}{4} = 1.5$

34. 00001.60

Sol.



y = 8

so value of $\frac{y}{5} = 1.6$

35. 00000.25

Sol. B is incorrect

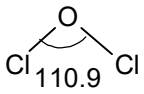
$AlCl_3 + 2ICl(melt) \longrightarrow [I_2Cl]^+ + [AlCl_4]^-$

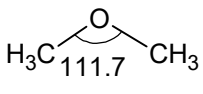
A, C, D and E are correct.

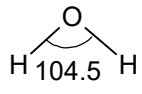
$\frac{\text{number of incorrect reaction}}{\text{number of correct reaction}} = \frac{1}{4} = 0.25$

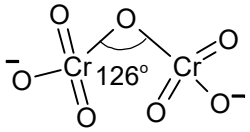
36. 00002.50

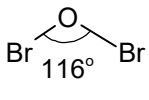
Sol.

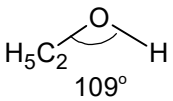


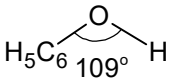


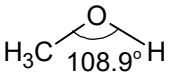


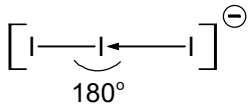












Total number of compound which have bond angle greater than $109^{o}\,28'$ are x = 5.

So, $\frac{x}{2} = \frac{5}{2} \Rightarrow 2.5$

Mathematics

PART – III

SECTION – A

37. A, B, C, D

Sol. $\beta = 2\alpha + 3$ and $4 \le |\alpha - \beta + 1| < 6 \Rightarrow \alpha \in (-8, -6] \cup [2, 4) \Rightarrow \alpha = \{-7, -6, 2, 3\}$

38. A, C

Sol. Point M can be $\left(\frac{3}{\sqrt{5}}, \frac{4}{\sqrt{5}}\right)$ or $\left(\frac{-3}{\sqrt{5}}, \frac{-4}{\sqrt{5}}\right)$

$\Rightarrow$ Equation of tangent is $x + 3y - 3\sqrt{5} = 0$ or $x + 3y + 3\sqrt{5} = 0$

39. B, C, D

Sol. Locus is hyperbola with foci (4, 2) and (–4, –2) and length of transverse axis = 4

40. A, B, C

Sol. Locus is $x^2 = \frac{4}{3}(y - 3)$

41. A, B, C, D

Sol. $\left(1 + x + x^2 + x^3\right)^{100} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{300} x^{300}$

Put $x = 1 \Rightarrow a_0 + a_1 + \ldots\ldots + a_{300} = 4^{100} = (1024)^{20}$

Put $x = -1 \Rightarrow a_0 - a_1 + a_2 \ldots\ldots + a_{300} = 0$

Put $x = \frac{1}{x} \Rightarrow a_r - a_{300-r} = 0$

By differentiating both sides and put x = 0, we get $a_1 = 100$

42. B, C

Sol. $\alpha = 5\pi - 15$, $\beta = 2\pi - 5 \Rightarrow \alpha - 3\beta = -\pi \Rightarrow \sin^2 \alpha + \cos^2 3\beta = 1$

43. A, B, C, D

Sol. Vector parallel to the line of intersection of planes $\vec{r} \cdot \vec{n}_1 = q_1$, $\vec{r} \cdot \vec{n}_2 = q_2$ is $\left(\vec{n}_1 \times \vec{n}_2\right)$

44. A, B, C, D

Sol. The lines bisecting the angles between the line $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z-6}{6}$ and $\frac{x-2}{3} = \frac{y-3}{6} = \frac{z-6}{2}$ are $\frac{x-2}{5} = \frac{y-3}{9} = \frac{z-6}{8}$ and $\frac{x-2}{1} = \frac{y-3}{3} = \frac{z-6}{-4}$

45. D

46. B

Sol. **(for Q.45 – Q.46)**

(I) $f(x) = \cos 2x - 10 \cos x - 11 \Rightarrow f(x) = 2\cos^2 x - 10 \cos x - 12$
$\Rightarrow f(x) = 2[\cos^2 x - 5 \cos x - 6] \Rightarrow$ Range of f(x) is [–20, 0]

(II) Period of function = $\frac{\pi}{\pi/10} = 10$

(III) $m = \frac{\pi/2}{5\pi/2} \Rightarrow m = \frac{1}{5}$

(IV) f(x) = constant

47. D

48. C

Sol. **(for Q.47 – Q.48)**

(I) (x, y) can be (0, 0) only

(II) $4a = \frac{4 \times 6 \times 2}{6 + 2} = 6$

(III) (a, b) = (–2, –8) $\Rightarrow$ 2a – b = 4

(IV) $a \in R$

SECTION – C

49. 00003.00

Sol. $\vec{z} = \frac{8\hat{i} + 10\hat{j} + 19\hat{k}}{7}$

50. 00045.00

Sol. $(6\{x\})^2 = 2[x] \cdot 3x \Rightarrow x = \frac{3}{2} \Rightarrow S = \frac{9}{20} \Rightarrow 100S = 45$

51. 11332.20

Sol. (a – b)(a + b) = c $\Rightarrow$ a – b = 1 $\Rightarrow$ a = 3, b = 2 $\Rightarrow$ c = 5 $\Rightarrow$ d = 7

52. 00005.00

Sol. $\cos 2x = \cos^3 x - \sin^3 x$

$\Rightarrow \cos^2 x - \sin^2 x = \cos^3 x - \sin^3 x \Rightarrow \cos x = 1$ or $\sin x = 1$ or $\sin x = \cos x$

53. 00000.00

Sol. $\tan 70° - \tan 20° = 2 \tan 50° \Rightarrow z = 2y$

54. 00010.00

Sol. $|z|_{min} = \frac{3}{\sqrt{13}}$